

BIOL 250 · Axial Systemic — Neural Tissue · ANS · Tracts · Senses — PAPER NOTES

QUICK-SCAN INDEX (alphabetical — scan by eye)

Neural tissue + ANS + tracts + senses. Scan this page first.

TERM	QUICK FACT
Action potential	Depolarization (Na ⁺ rush in) → repolarization (K ⁺ rush out) → hyperpolarization → resting
Afferent	Sensory signals TO CNS
Astrocytes	Most abundant CNS glial; BBB maintenance + K ⁺ buffering
Axon hillock	Where action potential is generated (lowest threshold)
Blood-brain barrier (BBB)	Tight junctions of CNS capillary endothelium + astrocyte endfeet
Convergence	Many inputs → one neuron
Divergence	One neuron → many targets
Efferent	Motor signals FROM CNS → muscle/gland
Ependymal cells	Line ventricles; produce CSF
Gray matter (brain/cord)	Unmyelinated axons + cell bodies; cortex + deep nuclei + horns
IPSP	Inhibitory PSP; hyperpolarizes membrane → harder to fire
Microglia	CNS macrophages; phagocytize debris + pathogens
Multiple sclerosis	Demyelinating disease → slowed/blocked conduction
Myelin	Schwann cells (PNS) or oligodendrocytes (CNS); speeds saltatory conduction
Nodes of Ranvier	Gaps in myelin; AP jumps between (saltatory conduction)
Oligodendrocytes	CNS myelin; 1 cell can myelinate MULTIPLE axons
Refractory (absolute)	Na ⁺ channels inactivated; impossible to fire; determines max firing rate
Refractory (relative)	K ⁺ channels open; can fire with STRONGER stimulus
Saltatory conduction	AP jumps node to node; faster + less energy
Satellite cells	PNS support cells surrounding neuron cell bodies in ganglia
Schwann cells	PNS myelin (1 cell = 1 segment of 1 axon)
Spatial summation	Multiple simultaneous inputs summed at axon hillock
Temporal summation	Rapid successive inputs from same neuron summed
White matter (brain/cord)	Myelinated axons; deep tracts + commissures + funiculi
Alpha-1 receptors	Smooth muscle contraction (vasoconstriction, mydriasis); NE
Beta-1 receptors	Heart ↑ rate + contractility; NE/Epi; blocked by metoprolol
Beta-2 receptors	Bronchodilation + vasodilation skeletal muscle; Epi
Cholinergic synapse	Uses ACh; parasympathetic + all preganglionic + somatic NMJ
Dual innervation	Most organs get BOTH sympathetic + parasympathetic
Fight-or-flight	Sympathetic: ↑HR, ↑BP, bronchodilation, ↓digestion, pupil dilation
Muscarinic receptors	ACh-binding G-protein coupled; parasympathetic target organs
Nicotinic receptors	ACh-binding ion channels; all ganglia + NMJ
Norepinephrine (NE)	Postganglionic sympathetic (except sweat glands = ACh)
Rest-and-digest	Parasympathetic: ↓HR, ↑digestion, bronchoconstriction, miosis, salivation
Sweat glands	Sympathetic BUT ACh (cholinergic) — exam trap!
Anterolateral (spinothalamic)	Pain + temperature; CROSS at spinal cord → contralateral cortex
Dorsal column-medial lemniscal	Fine touch + proprioception + vibration; CROSS at medulla
LMN signs	Flaccid paralysis + hyporeflexia + atrophy + fasciculations
UMN signs	Spastic paralysis + hyperreflexia + Babinski positive
Auditory pathway	Cochlear hair cells → CN VIII → cochlear nuclei → inferior colliculus → MGN thalamus → auditory cortex (temporal lobe)
Cochlea	Hearing organ; fluid-filled; hair cells in organ of Corti transduce sound
Cones	Color + daylight vision; concentrated at fovea centralis
Cornea	Outer transparent layer; most refraction of light
Fovea centralis	Point of sharpest vision; densest cone concentration
Gustation (taste)	CN VII (ant 2/3 tongue) + CN IX (post 1/3) + CN X (epiglottis)
Macula (utricle/saccule)	Static equilibrium; gravity + linear acceleration
Olfaction (smell)	CN I → olfactory bulb → limbic system + cortex; only sense bypassing thalamus
Optic disc (blind spot)	No photoreceptors; where optic nerve exits
Rods	Low-light/peripheral vision; more numerous; no color
Semicircular canals	Dynamic equilibrium (rotational); 3 planes; cristae ampullaris
Tympanic membrane	Eardrum; vibrates with sound → ossicles → oval window
Visual pathway	Retina → optic nerve → optic chiasm (nasal fibers cross) → LGN thalamus → occipital cortex

1. NEURAL TISSUE — Cell Types & Signaling

1.1 Glial Cells

CELL TYPE	LOC ATIO N	FUNCTION	EXAM KEY
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Astrocytes	CNS	BBB maintenance + K ⁺ buffering + structural support	MOST ABUNDANT glial cell
Oligodendrocytes	CNS	Myelin formation	1 cell → MULTIPLE axon segments
Microglia	CNS	CNS macrophages; phagocytize debris + pathogens	Immune surveillance
Ependymal cells	CNS	Line ventricles; produce + circulate CSF	Cilia help circulate CSF
Schwann cells	PNS	Myelin formation	1 cell = ONE axon segment; allow regeneration
Satellite cells	PNS	Surround neuron cell bodies in ganglia	Support/insulate

1.2 Action Potential Steps

STEP	EVENT	KEY ION
Resting (−70 mV)	Na ⁺ /K ⁺ ATPase maintains gradient; K ⁺ leaks out	Na ⁺ outside ↑, K ⁺ inside ↑
Threshold (−55 mV)	Sufficient depolarization → voltage-gated Na ⁺ channels OPEN	Na ⁺
Depolarization (rise)	Na ⁺ rushes IN; membrane → +30 mV	Na ⁺ IN
Repolarization (fall)	Na ⁺ channels INACTIVATE; K ⁺ channels OPEN; K ⁺ rushes OUT	K ⁺ OUT
Hyperpolarization	K ⁺ channels slow to close; briefly below −70 mV	K ⁺ OUT overshoot
Absolute refractory	Na ⁺ channels inactivated; IMPOSSIBLE to fire	Na ⁺ inactivated
Relative refractory	K ⁺ channels still open; CAN fire with STRONGER stimulus	K ⁺ OUT

AP generated at AXON HILLOCK. Propagates ONE direction only. Saltatory = jumps Nodes of Ranvier = faster + less ATP. MS destroys myelin → slowed conduction.

1.3 Synaptic Transmission & Summation

CONCEPT	DEFINITION / KEY FACT
Spatial summation	Multiple simultaneous inputs from DIFFERENT neurons summed at hillock
Temporal summation	Rapid successive inputs from SAME neuron summed before decay
EPSP	Excitatory PSP; depolarizes (Na ⁺ in)
IPSP	Inhibitory PSP; hyperpolarizes (K ⁺ out or Cl [−] in)
Convergence	Many neurons → 1 neuron; integration
Divergence	1 neuron → many neurons; amplification
Main CNS inhibitory NT	GABA (brain); Glycine (spinal cord)
Main CNS excitatory NT	Glutamate
NMJ neurotransmitter	ACh → nicotinic receptors on motor end plate
Nerve regeneration (PNS)	Schwann cells form tube; axon regrows ~1–4 mm/day; CNS cannot regenerate

2. AUTONOMIC NERVOUS SYSTEM (ANS)

△ GR traps: Sweat glands = sympathetic but ACh. Adrenal medulla = modified sympathetic ganglion (preganglionic ACh → Epi/NE). All preganglionic = ACh. Somatic NMJ = ACh (nicotinic).

2.1 Sympathetic vs. Parasympathetic

FEATURE	SYMPATHETIC	PARASYMPATHETIC
Origin	T1–L2 (thoracolumbar)	CN III,VII,IX,X + S2–S4 (craniosacral)
Preganglionic fiber	SHORT myelinated	LONG myelinated
Postganglionic fiber	LONG unmyelinated	SHORT unmyelinated
Ganglion location	Paravertebral chain (near cord)	Near/inside target organ
Preganglionic NT	ACh → nicotinic receptors	ACh → nicotinic receptors
Postganglionic NT	NE → adrenergic receptors	ACh → muscarinic receptors
Exceptions	Sweat glands = ACh; adrenal medulla = Epi	None
Heart	↑ HR + contractility (β1)	↓ HR (vagus / CN X)
Lungs	Bronchodilation (β2)	Bronchoconstriction
Pupils	Dilation — mydriasis (α1)	Constriction — miosis
Digestion	↓ peristalsis + sphincter contraction	↑ peristalsis + ↑ secretion
Bladder	Retention (sphincter contracts α1)	Voiding (detrusor contracts)
Sweat glands	Secretion — ACh (cholinergic!)	No innervation
Mnemonic	Fight or Flight	Rest and Digest

2.2 Receptors Quick Reference

RECEPTOR	AGO NIST	LOCATION / EFFECT	DRUG EXAMPLE
α1 (alpha-1)	NE > Epi	Blood vessels → vasoconstriction; iris → mydriasis	Phenylephrine (decongestant)

β1 (beta-1)	NE = Epi	Heart → ↑ HR + contractility	Metoprolol blocks β1 → ↓ HR
β2 (beta-2)	Epi > NE	Bronchial smooth muscle → relaxation (dilation)	Albuterol = β2 agonist
Nicotinic (Nn)	ACh	ALL autonomic ganglia	Hexamethonium blocks
Nicotinic (Nm)	ACh	NMJ (skeletal muscle)	Curare blocks → paralysis
Muscarinic (M)	ACh	Parasympathetic target organs → SLUDD	Atropine blocks → tachycardia + mydriasis

3. SENSORY & MOTOR TRACTS

3.1 Ascending (Sensory) Tracts

TRACT	MODALITY	CROSSES WHERE	DESTINATION
Dorsal column – medial lemniscus	Fine touch, vibration, proprioception	MEDULLA (nucleus gracilis/cuneatus)	Thalamus → somatosensory cortex (parietal lobe)
Anterolateral (spinothalamic)	Pain, temperature, crude touch	SPINAL CORD (at entry level)	Thalamus → somatosensory cortex
Spinocerebellar	Unconscious proprioception	Does NOT cross (mostly ipsilateral)	Cerebellum

Brown-Séquard (spinal hemisection): IPSILATERAL fine touch/motor loss + CONTRALATERAL pain/temp loss BELOW injury.

3.2 Descending (Motor) Tract

TRACT	FUNCTION	CROSSES WHERE
Corticospinal (pyramidal)	Voluntary skeletal muscle control	Medullary pyramids (85%); rest at cord level
Rubrospinal	Limb flexor control	Midbrain
Reticulospinal	Posture + locomotion + reflexes	Variable
Vestibulospinal	Balance + muscle tone	Mostly ipsilateral

3.3 UMN vs. LMN Lesion Signs

SIGN	UMN (brain/cord above ant. horn)	LMN (ant. horn → peripheral nerve)
Paralysis type	Spastic (↑ tone)	Flaccid (↓ tone)
Reflexes	Hyperreflexia	Hyporeflexia / absent
Babinski	POSITIVE (toes fan)	Negative
Atrophy	Mild (disuse)	Marked muscle wasting
Fasciculations	Absent	Present

4. SPECIAL & GENERAL SENSES

Also covered in HN_Notes Section 6. Duplicated here for the Systemic GR.

4.1 Hearing & Equilibrium

STRUCTURE	FUNCTION	EXAM KEY
Tympanic membrane (eardrum)	Vibrates with sound → ossicles	First structure to receive sound
Ossicles (malleus/incus/stapes)	Amplify + transmit vibration → oval window	Malleus = attached to eardrum; stapes = oval window
Oval window	Boundary ear → cochlea; stapes pushes it	Fluid wave into cochlea
Cochlea	Hearing; fluid-filled; organ of Corti has hair cells	Hair cells transduce fluid movement → AP
Organ of Corti	Receptor organ for hearing; hair cells on basilar membrane	CN VIII carries signal
Semicircular canals	Dynamic equilibrium (rotation); 3 planes	Cristae ampullaris = receptor; cupula displaced by fluid
Utricle + saccule	Static equilibrium; gravity + linear acceleration	Macula = receptor; otoliths bend hair cells
CN VIII (vestibulocochlear)	Both hearing (cochlear branch) + balance	Damage → deafness + vertigo

	(vestibular branch)	
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4.2 Vision

STRUCTURE	FUNCTION	EXAM KEY
Cornea	Transparent outer layer; MOST refraction	Not vascularized; nourished by aqueous humor
Lens	Fine-tune focus (accommodation)	Ciliary muscles change lens shape
Retina	Contains photoreceptors (rods + cones)	Inner layer; contains optic disc
Rods	Low-light + peripheral vision; no color	More numerous; distributed peripherally
Cones	Color + daylight vision	Concentrated at fovea centralis
Fovea centralis	Sharpest vision; densest cone packing	Center of macula lutea
Optic disc	Blind spot; no photoreceptors; optic nerve exits	Can see it with ophthalmoscope
Visual pathway	Retina → optic nerve → optic chiasm → LGN thalamus → occipital cortex	Nasal fibers cross at chiasm; temporal stay ipsilateral

4.3 Taste, Smell & General Senses

SENSE	RECEPTOR / CN	KEY FACT
Olfaction (smell)	Olfactory epithelium → CN I → olfactory bulb	ONLY sense bypassing thalamus → goes to limbic directly
Gustation (taste)	CN VII (ant 2/3 tongue) + CN IX (post 1/3) + CN X (epiglottis)	5 tastes: sweet, sour, salty, bitter, umami
Mechanoreceptors	Meissner (light touch) + Pacinian (pressure/vibration) + Merkel (sustained touch) + Ruffini (stretch)	Encapsulated = more sensitive + specific
Thermoreceptors	Free nerve endings (FNE); cold vs. warm fibers	Temp extremes activate nociceptors too
Nociceptors (pain)	Free nerve endings; Aδ fibers (fast/sharp) + C fibers (slow/dull)	C fibers = unmyelinated = slow burning pain
Proprioceptors	Muscle spindles (length) + Golgi tendon organs (tension) + joint receptors	Muscle spindle → stretch reflex; GTO → inhibition

RAPID-FIRE SYSTEMIC CHEAT SHEET

#	QUESTION	ANSWER
1	Most abundant CNS glial?	Astrocytes
2	CNS myelin → 1 cell covers?	MULTIPLE axon segments (oligodendrocytes)
3	PNS myelin → 1 cell covers?	ONE axon segment (Schwann cells)
4	CNS macrophages?	Microglia
5	CSF produced by?	Ependymal cells
6	AP generated at?	Axon hillock
7	Absolute refractory =?	Na ⁺ inactivated; impossible to fire
8	Saltatory conduction jumps?	Nodes of Ranvier
9	MS affects?	Myelin (oligodendrocytes in CNS)
10	Main CNS inhibitory NT?	GABA
11	Main CNS excitatory NT?	Glutamate
12	PNS can regenerate? CNS?	PNS yes (Schwann tube); CNS no
13	Sympathetic origin?	T1–L2 (thoracolumbar)
14	Parasympathetic origin?	CN III, VII, IX, X + S2–S4
15	ALL preganglionic NT?	ACh (nicotinic)
16	Sympathetic postganglionic NT?	NE (EXCEPT sweat glands = ACh)
17	Parasympathetic postganglionic NT?	ACh → muscarinic receptors
18	Sweat glands NT?	ACh — sympathetic but cholinergic!
19	β1 effect?	↑ heart rate + contractility
20	β2 effect?	Bronchodilation
21	α1 effect?	Vasoconstriction + mydriasis
22	Atropine blocks?	Muscarinic → ↑ HR + mydriasis
23	Pain/temp tract → crosses where?	Spinal cord (anterolateral)
24	Fine touch tract → crosses where?	Medulla (dorsal column)
25	UMN signs?	Spastic + hyperreflexia + Babinski+
26	LMN signs?	Flaccid + hyporeflexia + atrophy + fasciculations
27	Only sense bypassing thalamus?	Olfaction (CN I → limbic directly)
28	Taste CNs?	VII (ant 2/3) + IX (post 1/3) + X (epiglottis)
29	Fovea centralis =?	Sharpest vision; most cones
30	Optic disc =?	Blind spot; no photoreceptors
31	Nasal fibers cross at?	Optic chiasm
32	Cochlea → hearing organ; receptor cells?	Hair cells in organ of Corti
33	Semicircular canals =?	Dynamic equilibrium (rotation)
34	Utricle/sacculle =?	Static equilibrium (gravity + linear accel)

35	Muscle spindle detects?	Muscle LENGTH (stretch); triggers stretch reflex
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EXAM COVERAGE PATCH — TB Questions Not in Main Notes

All gaps from the Stuvia question bank. Format: KEY TERM / QUESTION STEM → ANSWER. Study these alongside the main sections.

1. Neural Tissue — Organization & Cell Details

KEY TERM / QUESTION STEM	ANSWER
ANS function (general)	Maintains homeostasis via visceral control
Most widely used ANS neurotransmitter	ACh (acetylcholine)
PNS functions	Provides sensory info TO CNS + carries motor commands FROM CNS
Receptor classification	Somatic and visceral
Afferent / sensory pathways	Pathways leading FROM receptors TO CNS
Information processing cells	Neurons
Gray matter interior clusters	Nuclei (larger groups of gray matter inside CNS)
Microglia — description	SMALLEST neuroglial cell; CNS macrophage
Satellite cells location	Near cell bodies in PNS ganglia (not in CNS)
Elevated microglia indicates	Infected or injured CNS tissue
Neurolemma	Outermost covering of a myelinated axon (formed by Schwann cell cytoplasm)
Nissl bodies / coarse grainy appearance	Rough ER + ribosomes; smooth ER does NOT contribute
Axon hillock	Connects cell body to axon; site of AP initiation
Neuron structural classification based on	Number of processes projecting from cell body
Neuron functional classifications	Sensory (afferent), motor (efferent), interneuron
Axoplasmic transport	Movement of nutrients, wastes, organelles between cell body ↔ axon terminals
Neurons cannot divide because	They lose centrioles during maturation → cannot undergo mitosis
Somatic sensory neurons associated with	Exteroceptors (detect external stimuli)
Nerve impulse conduction rate depends on	Presence or absence of myelin sheath (myelinated = faster)
Wallerian degeneration	Macrophages remove debris of damaged axons distal to injury
Reverberation	Collateral axons extend BACK toward impulse source → continued stimulation of original neuron

2. Synaptic Communication — Details

KEY TERM / QUESTION STEM	ANSWER
Most well-known neurotransmitter	Acetylcholine (ACh)
Nerve impulse at terminal bouton triggers	Release of ACh at presynaptic membrane
Nonvesicular synapse unique feature	Communicating junction (gap junction); no vesicles
Synapse function	Transfers information to another neuron OR to effector cell
Vesicular synapse communication	By neurotransmitters released from vesicles
Neurotransmitters stored in	Synaptic vesicles
Types of synapses	Vesicular + nonvesicular + electrical (all of the above)
Neuron temporarily cannot transmit if	Neurotransmitter supply is exhausted
Neuron pool	Group of interconnected neurons with specific functions

3. ANS — Sympathetic Division Details

KEY TERM / QUESTION STEM	ANSWER
Sympathetic division general function	Increases activity levels; fight-or-flight; ALL of the above
Gray ramus fiber type	Unmyelinated postganglionic (ALL of the above = sympathetic fibers)
Signs of sympathetic stimulation	Increased cardiovascular AND respiratory activity
Stellate ganglion	Fused cervical + thoracic chain ganglia
Suprarenal (adrenal) medulla	Highly specialized sympathetic region; releases Epi/NE into bloodstream → widespread response
Sympathetic actions	Accelerates heart rate + increases cardiac contraction strength
Alternate names for sympathetic chain	Lateral ganglia (also: paravertebral ganglia, vertebral chain)
Unpaired ganglia of ANS	Collateral ganglia (most ANS ganglia are paired)
Splanchnic nerves	Preganglionic fibers connecting sympathetic chain → collateral ganglia
Sacral splanchnic nerves end in	Hypogastric plexus
NE duration of action	Only a few seconds (brief effect at target)
Adrenal medulla Epi/NE duration	Relatively LONG (inactivated by enzymes in plasma; circulates in blood)
Neuroeffector synapse varicosities	Large branching terminal endings of postganglionic sympathetic neuron
Sympathetic division can release	NE, Epi, AND other substances (all of the above)
Middle/inferior cervical ganglia mediate	Increases in cardiac and respiratory rates
Sympathetic → respiratory effect	Increases respiratory activities needed for exercise

4. ANS — Parasympathetic Division Details

KEY TERM / QUESTION STEM	ANSWER
Cranial nerves with parasympathetic fibers	CN III, VII, IX, X

Parasympathetic ganglia in head (4)	Ciliary · Pterygopalatine · Submandibular · Otic
Sacral parasympathetic pathway	Forms distinct pelvic nerves → innervate intramural ganglia
Inferior colon/rectum control	All of the above (sacral parasympathetic)
Eye focusing (accommodation) pathway	Ciliary ganglion → ciliary muscle → lens shape change
Solar plexus =	Celiac plexus (sympathetic collateral ganglion)

5. ANS — Relationships, CNS Control & Reflexes

KEY TERM / QUESTION STEM	ANSWER
Headquarters of all autonomic function	Hypothalamus
ANS activity levels controlled by	Specific centers in the brainstem
Sympathetic + parasympathetic reflexes coordinated by	All of the above (multiple brain centers)
Vessel diameter = smallest when	Decreased sympathetic stimulation (less tone → vasoconstriction? No: INCREASED symp → vasoconstriction; DECREASED symp → less constriction = larger. Answer: decreased sympathetic stimulation allows vessel to dilate actually... wait — the answer is decreased sympathetic stimulation leads to SMALLEST diameter? Let me re-read: "decreased sympathetic stimulation" — vessels constrict = smallest. Actually parasympathetic dominant = dilate. Decreased sympathetic = dilate. This is a tricky one. The correct answer from TB: decreased sympathetic stimulation.)
Carotid artery BP rise triggers	All of the above (baroreceptor reflex responses)
Reflexes that respond to stretch	All of the above
Visceral reflexes	Provide autonomic motor responses; can be modified/facilitated/inhibited by higher centers
Brain regions affecting ANS regulation	All of the above (hypothalamus, limbic, cortex, brainstem)
Raynaud's phenomenon (finger/toe vasoconstriction)	Both local + systemic defects in sympathetic control of vessel tone
Megacolon cause	Reduction of parasympathetic innervation → excess dilation proximal to defect
Burst of energy from ANS	Sympathetic division increases glucose release from liver
Plasmalemmae (cell membranes)	All respond to changes in extracellular environment
Postganglionic sympathetic fibers length vs. parasympathetic	Sympathetic postganglionic = LONGER than parasympathetic postganglionic

6. Sensory & Motor Tracts — Specific Details

KEY TERM / QUESTION STEM	ANSWER
Third-order neurons	Extend FROM thalamus → THROUGH internal capsule → TO somatosensory cortex
Angina referred pain location	Upper left chest + arm; referred to corresponding primary sensory cortex region
Afferent =	Sensory
Sensation vs. perception	Sensation = information FROM receptor detecting stimulus (raw signal)
Corticobulbar tract	Upper motor neurons to lateral rectus (eye); controls eye movements via cranial nerve nuclei
Corticobulbar also controls	Motor nuclei of cranial nerves controlling eye movements
Tectospinal tract	Startle response (blinking/flinching at bright lights or loud sounds)
Corticobulbar damage	Inability to control voluntary eye/facial movements
Nuclei correctly paired with tracts	All of the above
Flaccid paralysis after sports injury	Lower motor neuron damaged (not UMN)
Sensory homunculus	Map showing number of sensory receptors in body regions
Distinguish sensations from different body areas	Sensory neurons from each region synapse in SPECIFIC brain regions
Complex motor patterns area	Cerebellum
Mesencephalon (midbrain) controls	Motor responses to visual AND auditory stimuli
Conscious thought + intellectual functions	Cerebral hemispheres

7. Receptors — Classification & Properties

KEY TERM / QUESTION STEM	ANSWER
Receptor characteristic	Each has a characteristic sensitivity (responds to specific stimulus)
Special sensory receptors differ from simple	Restricted to stimuli to which they respond
Larger receptive field →	POORER ability to localize stimulus
Can't detect EM waves (except visible light)	No receptors exist in body for those wavelengths
Tonic receptors	Always active; examples of constantly firing receptors
Sensory adaptation	Decline in activity along afferent fiber after initial strong response
Phasic + tonic combined receptors	Convey extremely complicated sensory information to CNS
Referred pain	Visceral pain felt in superficial areas (e.g. heart attack → left arm)

8. General Senses — Specific Facts

KEY TERM / QUESTION STEM	ANSWER
Unencapsulated receptors	Free nerve endings + root hair plexuses + tactile discs

Thermoreceptors	Cold receptors 3–4× more common than warm receptors
Nociceptors (pain receptors)	Generally have LARGE receptive fields
Deep pressure receptor → poor localization	Pacinian corpuscles (deep pressure receptors)
Blood pressure monitored by	All of the above (baroreceptors in multiple locations)
Blood oxygen monitored by	Aortic body (chemoreceptor near aortic arch)
Proprioceptors → CNS information	Position of structures (body position/movement)
Light touch to hair (cat brushing by)	Free nerve ending of hair root (root hair plexus)
Fingertips/toes lose heat faster	Greater surface area to volume ratio → lose heat faster

9. Reflexes — Additional Details

KEY TERM / QUESTION STEM	ANSWER
Spinal reflexes types	Include BOTH monosynaptic AND polysynaptic
Stretch reflex example	Knee jerk (patellar reflex)
Reflex motor response	All of the above (involuntary, rapid, stereotyped)
Higher brain centers + reflexes	Can ENHANCE or SUPPRESS spinal reflexes via descending tracts
Triceps jerk tests	Spinal segment C6–C7; test for damage to those levels

10. Special Senses — Equilibrium & Hearing Details

KEY TERM / QUESTION STEM	ANSWER
3 anatomical regions of ear	External ear · Middle ear · Inner ear
Membranous labyrinth	System of fluid-filled tubes + chambers of inner ear
Vestibular complex fluid	Endolymph (fills membranous labyrinth)
Cupula	Gelatinous mass in semicircular ducts; kinocilia + stereocilia embedded in it
Statoconia	Calcium carbonate crystals in utricle + saccule; bend hair cells with gravity
Auditory cortex damage	All aspects of hearing lost (all of the above)

11. Special Senses — Taste & Smell Details

KEY TERM / QUESTION STEM	ANSWER
Taste perception factors	All of the above (smell + texture + temperature + taste)
Most taste buds in adults	Circumvallate (vallate) papillae — largest, posterior tongue
Taste buds receptor type	Chemoreceptors
Olfactory receptors unique because	Rare example of neuronal replacement (new neurons generated throughout life)
Air swirled in nose to	Bring airborne compounds to olfactory organs (turbulence = better detection)
Older individuals + more perfume	Decrease in NUMBER of olfactory receptors with age

12. Special Senses — Vision Details

KEY TERM / QUESTION STEM	ANSWER
Cornea	Anteriorly curving bulge; where contact lenses sit; most refraction
Tear drainage pathway order	Lacrimal gland → surface of eye → lacrimal puncta → lacrimal sac → nasolacrimal duct → nasal cavity
Visual info first arrives at	Lateral geniculate nuclei (LGN) of thalamus
Endocrine + nervous systems together	Monitor and adjust physiological activities at any given moment

13. Cross-Sectional Anatomy — Key Landmarks

KEY TERM / QUESTION STEM	ANSWER
Midline structure at cervical cross-section	Nose (at that level)
Largest structure on lateral portion at mid-thoracic	Lungs
Most anterior structure at lumbar cross-section	Rectus abdominis muscles
Nervous system functions	All of the above (sensory, motor, integrative)

TB HOT SPOTS — SYSTEMIC / ANS / SENSES

Facts from the Stuvia TB that need quick reference — not fully covered above.

SYSTEMIC / ANS / SENSES

TERM / CONCEPT	KEY FACT (answer the TB is looking for)
Action potential mechanism	ABRUPT change in transmembrane potential; Na ⁺ rushes in (depolarization)
Esophageal achalasia	Excess PARASYMPATHETIC activity → excessive + irregular esophageal

	contraction
Hair root free nerve endings	Detect LIGHT TOUCH (e.g., cat brushing); surround hair follicle base
Astrocytes & BBB	Maintain BBB; if drug bypasses BBB → indicates astrocyte damage
Satellite cells	PNS glia; surround cell bodies IN GANGLIA; support + regulate
Microglia	CNS macrophages; elevated count in INFECTION or INJURY; biopsy shows increase
Demyelination	Gradual REDUCTION in myelin sheath; slows conduction
Neurons: no centrioles	CANNOT divide; rendered incapable of mitosis
Synapse sequence	Depol → Ca^{2+} entry → vesicle fusion → NT release → receptor binding → response (3 → 1 → 5 → 2 → 4)
Sensory/afferent neuron	Delivers information TO the CNS (NOT motor); classified by direction
CNS tract	Bundle of axons sharing same ORIGIN, DESTINATION + FUNCTION; true = tract
PNS nerve	Bundle of axons in PNS (with epi/peri/endoneurium wrappings)
Corticobulbar tract	Controls voluntary movement via cranial nerves; damage → poor control of facial/jaw mm.
Flaccid paralysis	LOWER motor neuron (LMN) damage; no tone, no reflexes, muscle atrophy
Spastic paralysis	UPPER motor neuron (UMN) damage; hyperreflexia, increased tone
Dual innervation	Vital organs receive BOTH sympathetic + parasympathetic input simultaneously
Raynaud's disease	Local + systemic defects in SYMPATHETIC response → vasoconstriction in fingertips/toes
Megacolon	REDUCTION in parasympathetic innervation → excess colon dilation proximal to defect
ANS overview	Involuntary; 2-neuron chain; pre-ganglionic + post-ganglionic; all correct = E
Cross-section: orientation	TOP of image = ANTERIOR; bottom = posterior; right/left flipped
Lungs in cross-section	LARGEST + most easily recognized structure on LATERAL portion
Uterus in cross-section	Visible in SACRAL LEVEL cross-sections (female pelvis)
Normal aging effects	All correct: slower conduction, weaker reflexes, neuron loss, etc.
Nociceptors location	Especially common in VISCERA (detect pain from internal organs)
Free nerve endings (hair root)	Detect light touch; respond to hair deflection
Thermoreceptors	Structural differences exist between cold + warm thermoreceptors
Taste buds	Chemoreceptors; respond to dissolved chemicals